ToxPipe Evaluation

ABT Question OCR and Dataset generation

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Introduction

Questions from American Board of Toxicology (ABT) certification exams (2000-20007) were extracted. For a few exams (2005-2007), the answers were provided in tabulated format within a non-pdf file (e.g. MS Excel and MS Word). For the rest, both the questions and answers were provided as scanned pdf format. These pdf files were processed using Amazon Textract which has OCR capability and "Azure GPT-4o" model.

Processing steps of the pdf files

The questions and the answers were processed using the following steps:

Step 1:

- The pdf files were uploaded to Amazon S3, as Amazon Textract does not allow "multipage" processing for on-premise files
- The relevant pages with questions and answers were filtered and preprocessed using regular expressions
- GPT-4o were used to list the questions and answers (both in string and tabular format) using separate prompts
- Each pdf file was processed separately

  Command

  ```
  python abt-question-ocr/src/extraction_questions.py
  ```

Step 2:

- The pdf outputs were combined with the non-pdf data

  Notebook

  ```
  abt-question-ocr/notebooks/combine_pdfs.ipynb
  ```

Output format

The output file contains the following columns:

- *Question Index*: Consists of exam year, exam section (or part) and question number. For example, "2000A Q1" denotes the question 1 of the part A of the examination year 2000.
- *Questions*: The question extracted as it was in the pdf
- *Options*: The options extracted as they were in the pdf
- *Question Source*: The source pdf file name that contains the question
- *Answers*: The answer choice(s) for the question
- *Answer Source*: The source pdf file name that contains the answer